This means, for a given magnitude of $\vec{E}$- field and $\vec{B}$- field, the forces act only on the particle moving with particular speed $v = \frac{E}{B}$. This speed is independent of mass and charge.

By proper choice of electric and magnetic fields, the particle with particular speed can be selected. Such an arrangement of fields is called a **velocity selector**.

EXAMPLE 3.22

Let E be the electric field of magnitude 6.0×10^6 N C^{-1} and B be the magnetic field magnitude 0.83 T. Suppose an electron is accelerated with a potential of 200 V, will it show zero deflection?. If not, at what potential will it show zero deflection.

Solution:

Electric field, $E = 6.0 \times 10^6$ N C^{-1} and magnetic field, $B = 0.83$ T.

Then

$$v = \frac{E}{B} = \frac{6.0 \times 10^6}{0.83}$$
$$= 7.23 \times 10^6 \text{ ms}^{-1}$$

When an electron goes with this velocity, it shows null deflection. Since the accelerating potential is 200 V, the electron acquires kinetic energy because of this accelerating potential. Hence,

$$\frac{1}{2}mv^2 = eV$$
$$v = \sqrt{\frac{2eV}{m}}$$

Since the mass of the electron, $m = 9.1 \times 10^{-31}$ kg and charge of an electron, $|q| = e = 1.6 \times 10^{-19}$ C. The velocity acquired by the electron due to accelerating potential of 200 V is

$$v_{200} = \sqrt{\frac{2\left(1.6 \times 10^{-19}\right)(200)}{\left(9.1 \times 10^{-31}\right)}} = 8.39 \times 10^6 \text{ m s}^{-1}$$

Since the speed $v_{200} > v$, the electron is deflected towards direction of Lorentz force. So, in order to have null deflection, the potential, we have to supply is

$$V = \frac{1}{2}\frac{mv^2}{e} = \frac{\left(9.1 \times 10^{-31}\right) \times \left(7.23 \times 10^6\right)^2}{2 \times \left(1.6 \times 10^{-19}\right)}$$
$$V = 148.65 \text{ V}$$

3.10.4 Cyclotron

Cyclotron (Figure 3.49) is a device used to accelerate the charged particles to gain large kinetic energy. It is also called as high energy accelerator. It was invented by Lawrence and Livingston in 1934.

Principle

When a charged particle moves perpendicular to the magnetic field, it experiences magnetic Lorentz force.

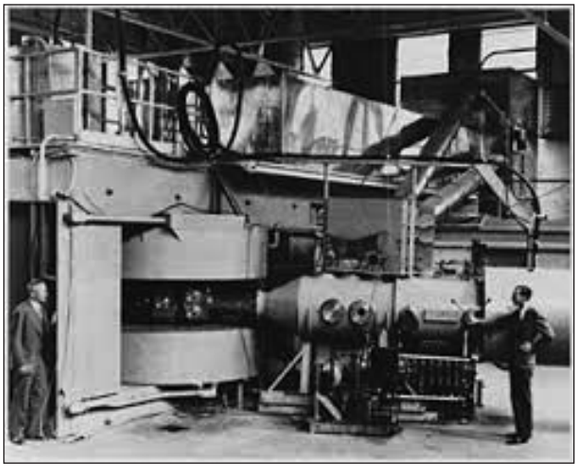


Figure 3.49 Cyclotron invented by Lawrence and Livingston

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